

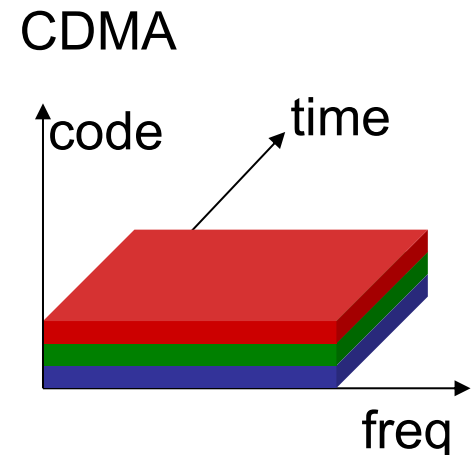
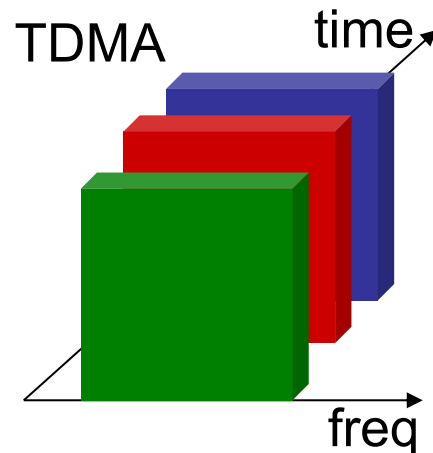
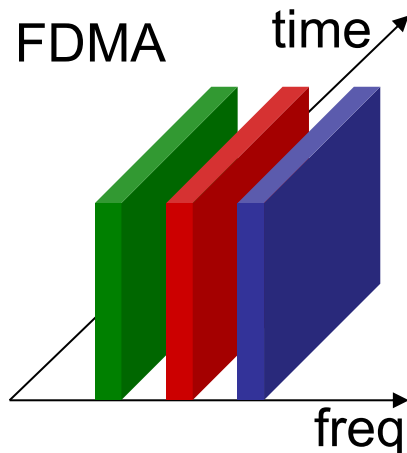
The Link-Layer

- The IoT is made by heterogeneous links interconnected at the network (IP) layer
- Key link-layer features:
 - Framing
 - Addressing
 - Error checking
 - Length indication
 - Broadcast and unicast

HTTP, DNS, XMPP, MQTT, CoAp, TLS, SSL	Application layer
TCP, UDP	Transport Layer
IP(V4, V6), 6LowPAN	Internet Layer
Ethernet, 802.11 WiFi, 802.15.4	Link Layer
Electric, optical or RF transmission	Physical Layer

Medium Access Control

- The sharing of a radio by multiple independent devices
- There are multiple ways to share a radio
 - Frequency Division Multiple Access (FDMA)
 - Time Division Multiple Access (TDMA)
 - Carrier Sense Multiple Access (CSMA)
 - Code Division Multiple Access (CDMA)
 - Hybrids of the above

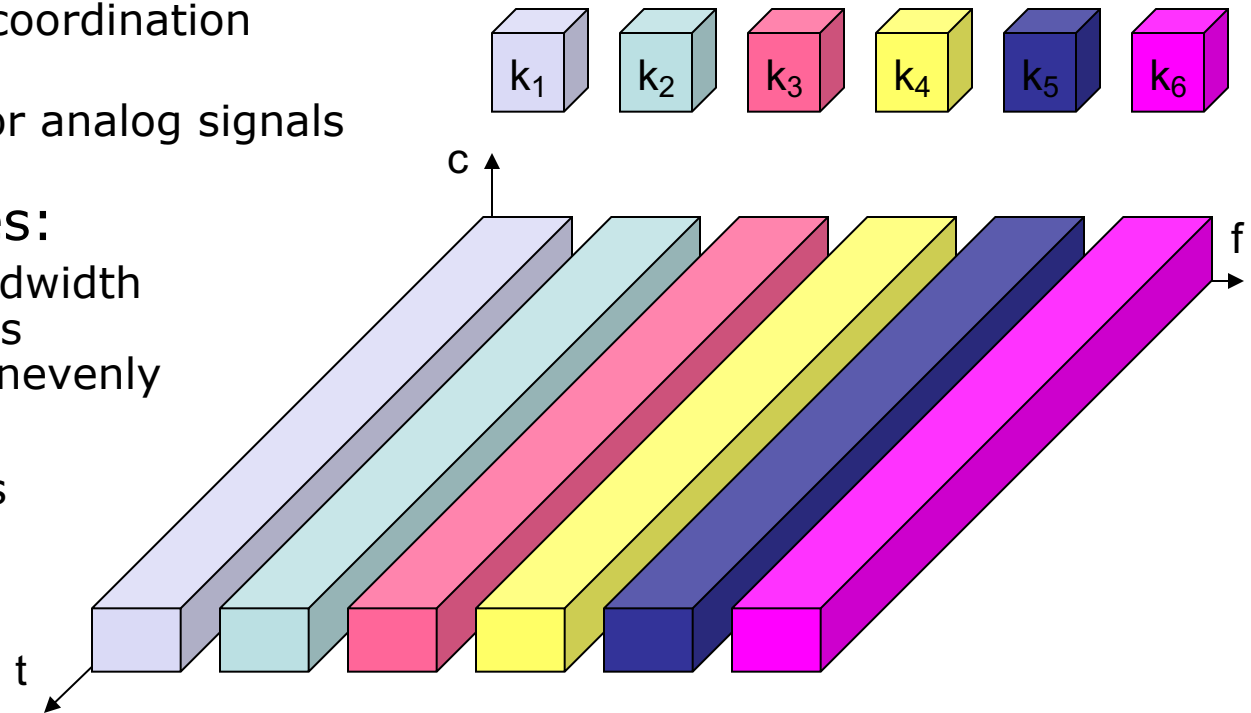


Medium Access Control

- MAC algorithms also take care of
 - Acknowledgements for packets
 - Link topology and addressing
 - Error checking and link security

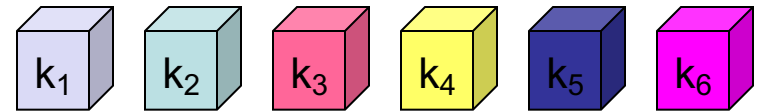
FDMA

- Separation of the spectrum into smaller frequency bands
- A channel gets a certain band of the spectrum for the whole time
- Advantages:
 - no dynamic coordination necessary
 - works also for analog signals
- Disadvantages:
 - waste of bandwidth if the traffic is distributed unevenly
 - inflexible
 - guard spaces



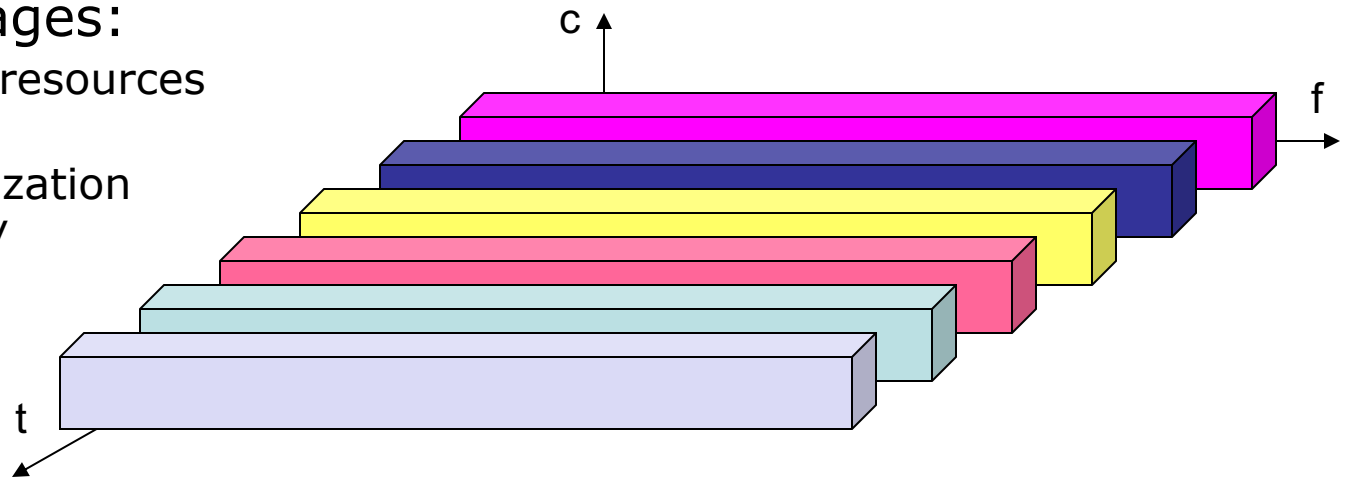
TDMA

- A channel gets the whole spectrum for a certain amount of time
- Advantages:
 - only one carrier in the medium at any time
 - throughput high even for many users



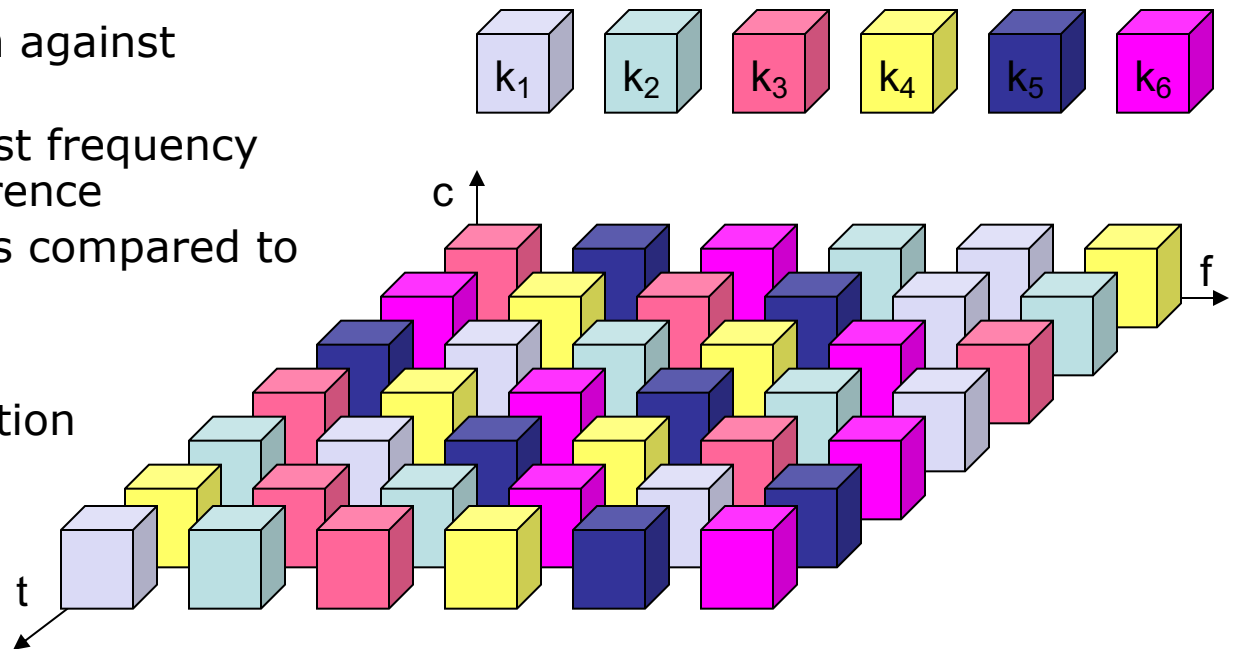
- Disadvantages:

- Waste of resources
- precise synchronization necessary



Time and frequency multiplex

- Combination of both methods
- A channel gets a certain frequency band for a certain amount of time
 - Example: GSM
- Advantages:
 - better protection against tapping
 - protection against frequency selective interference
 - higher data rates compared to code multiplex
- but:
 - precise coordination required



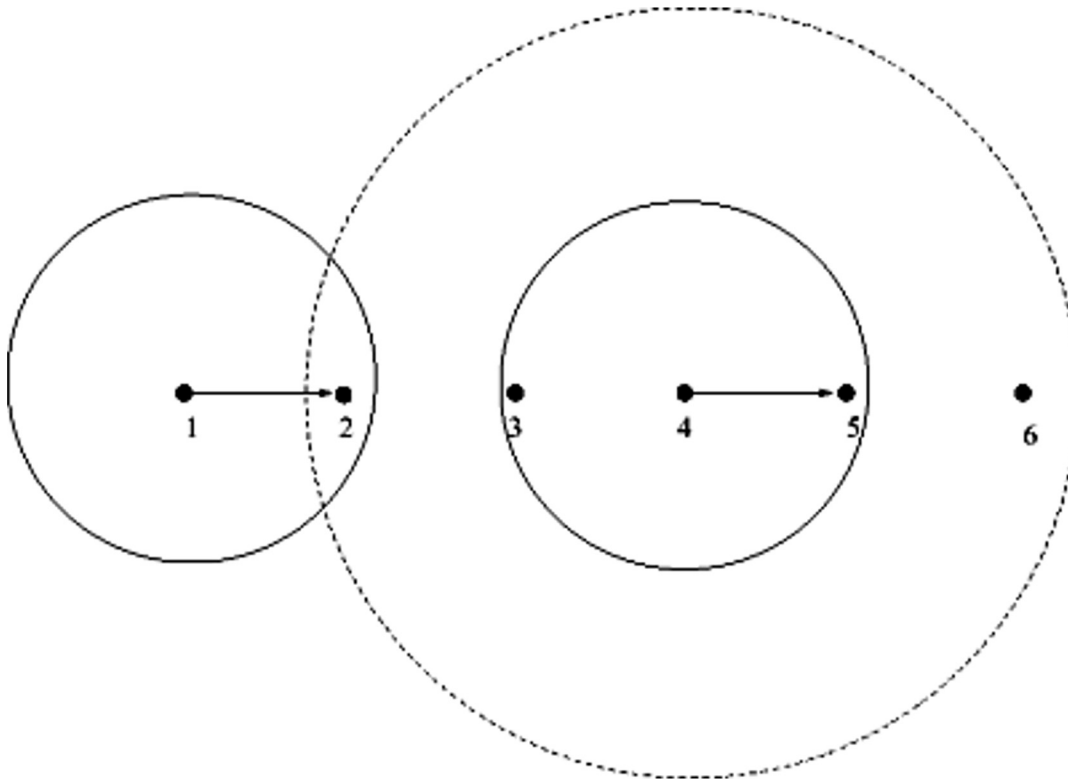
Carrier Sense Multiple Access (CSMA)

- Listen before you speak
- Check whether the medium is active before sending a packet (i.e. *carrier sensing*)
- If medium idle, then transmit
- If collision happens, then detect and resolve
- If medium is found busy, transmission follows:
 - 1- persistent
 - P- persistent
 - Non-persistent

Collision detection (CSMA/CD)

- All aforementioned scheme can suffer from collision
- Device can detect collision
 - Listen while transmitting
 - Wait for $2 * \text{propagation delay}$
- On collision detection wait for random time before retrying
- Binary Exponential Backoff Algorithm
 - Reduces the chances of two waiting stations picking the same random time

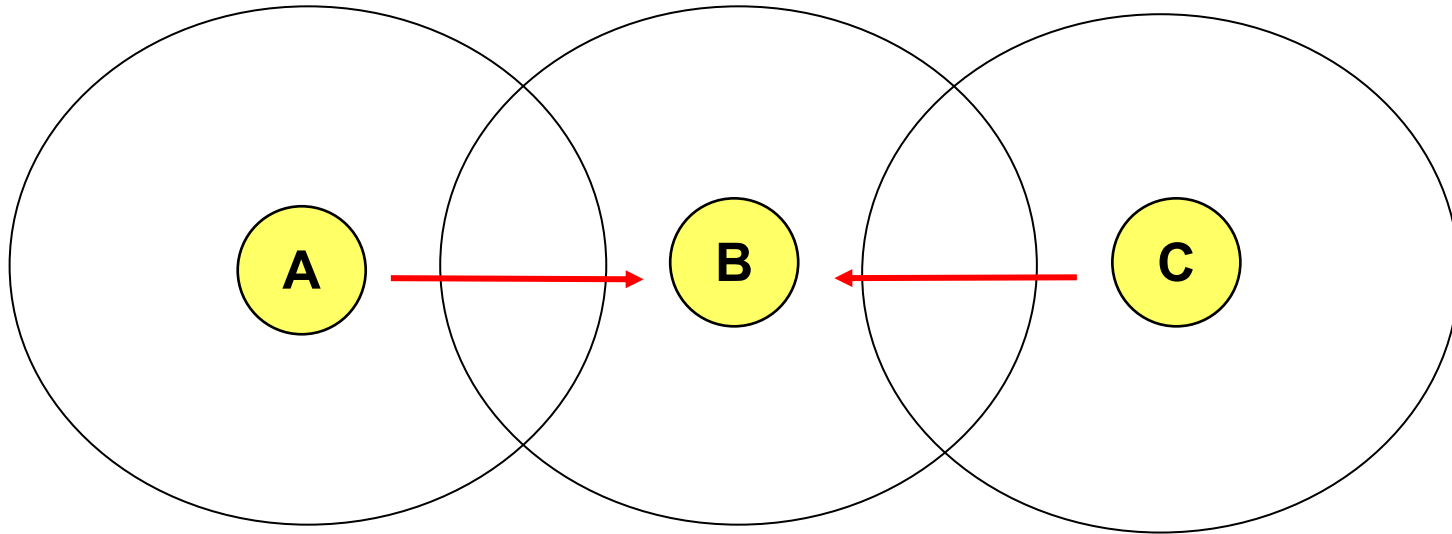
Effect of interference and coverage range



Transmission from 1 $\Rightarrow$ 2
will fail when 4 is
transmitting to 5 due to
interference

Transmission from 1 $\Rightarrow$ 3 or 4,5,6 is impossible (out of coverage)

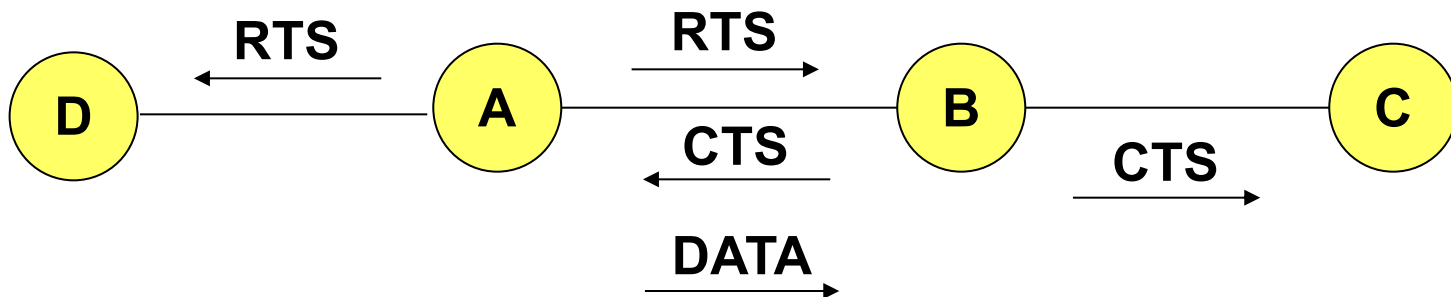
Hidden Terminal Problem



- A and C cannot hear each other.
- A sends to B, C cannot receive A.
- C wants to send to B, C senses a “free” medium (**CS fails**)
- Collision occurs at B.
- A cannot receive the collision (**CD fails**).
- A is “hidden” for C.

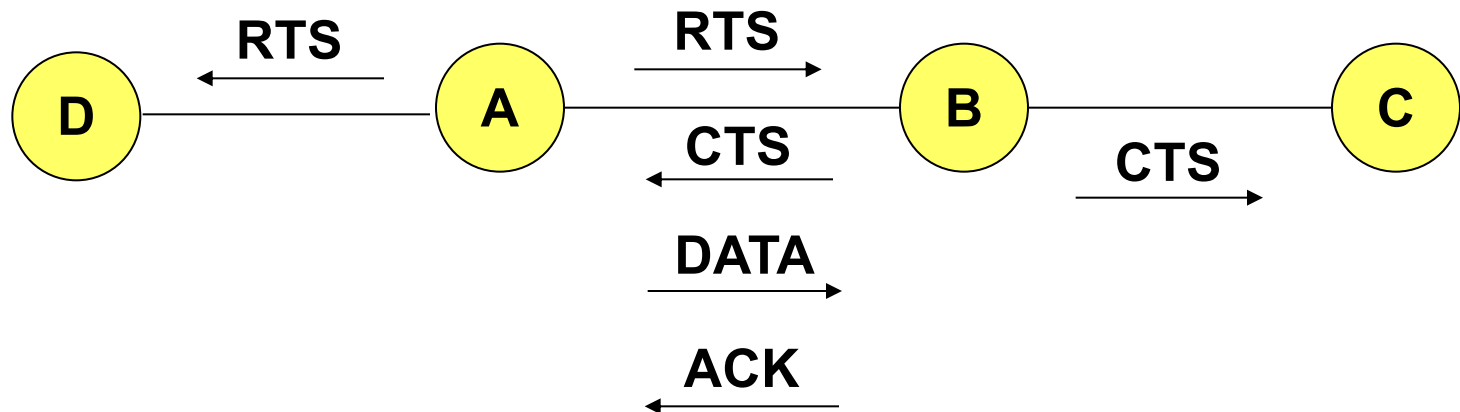
Solution for Hidden Terminals

- A first sends a *Request-to-Send (RTS)* to B
- On receiving **RTS**, B responds *Clear-to-Send (CTS)*
- Hidden node C overhears **CTS** and keeps quiet
 - Transfer duration is included in both RTS and CTS
- Exposed node overhears a **RTS** but not the **CTS**
 - D's transmission cannot interfere at B

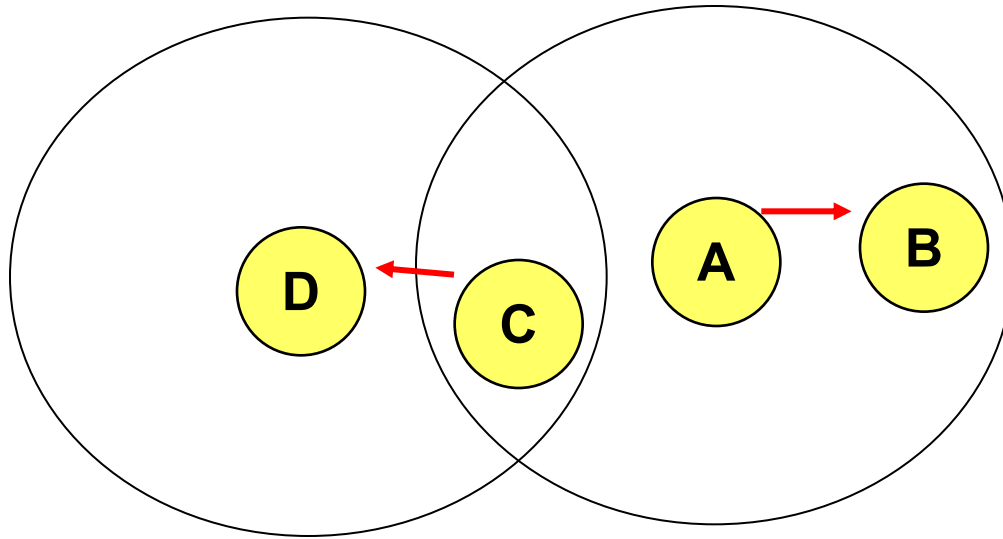


Adding ACKs

- Use **acknowledgements**
 - When B receives DATA from A, B sends an **ACK**
 - If A fails to receive an **ACK**, A retransmits the DATA
 - Both C **and** D remain quiet until **ACK** (to prevent collision of **ACK**)
 - Expected duration of transmission+ACK is included in **RTS/CTS** packets



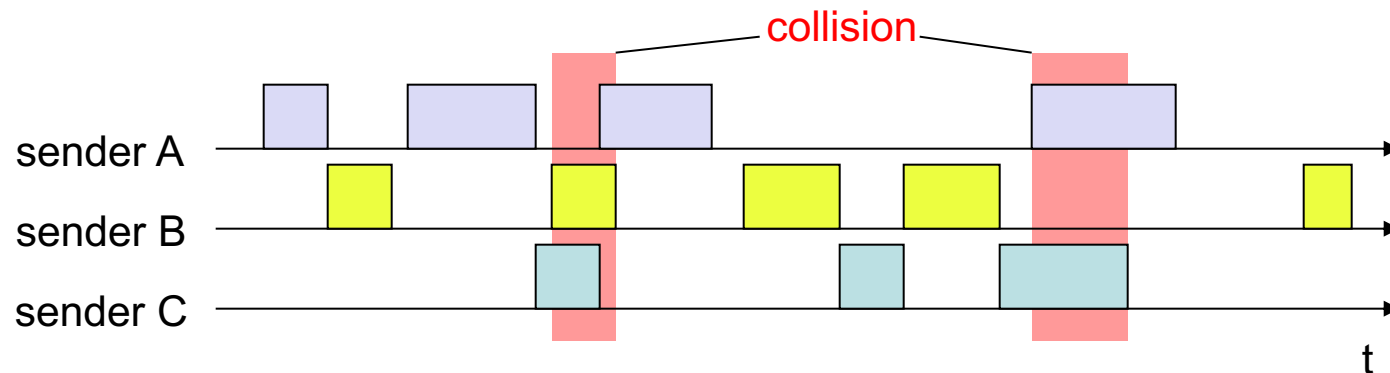
Exposed Terminal Problem



- A starts sending to B.
- C senses carrier, finds medium in use and has to wait for A->B to end.
- D is outside the range of A, therefore waiting is not necessary.
- A and C are “exposed” terminals

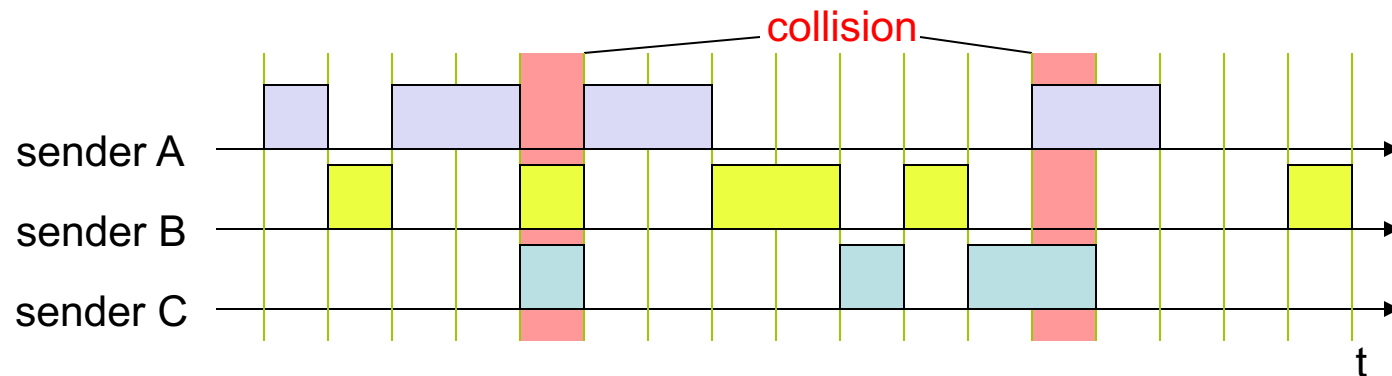
Aloha

- Random access scheme, Time division Multiplexed
- No central arbiter controlling access
- No coordination among the stations
- If two or more stations transmit at the same time:
 - a collision occurs
 - the transmitted data is destroyed.
- Stations transmit whenever they have data to send
 - Detect collision or wait for acknowledgment
 - If no ack (or collision), try again after a random waiting time
 - If there is a collision, all nodes have to re-transmit packets



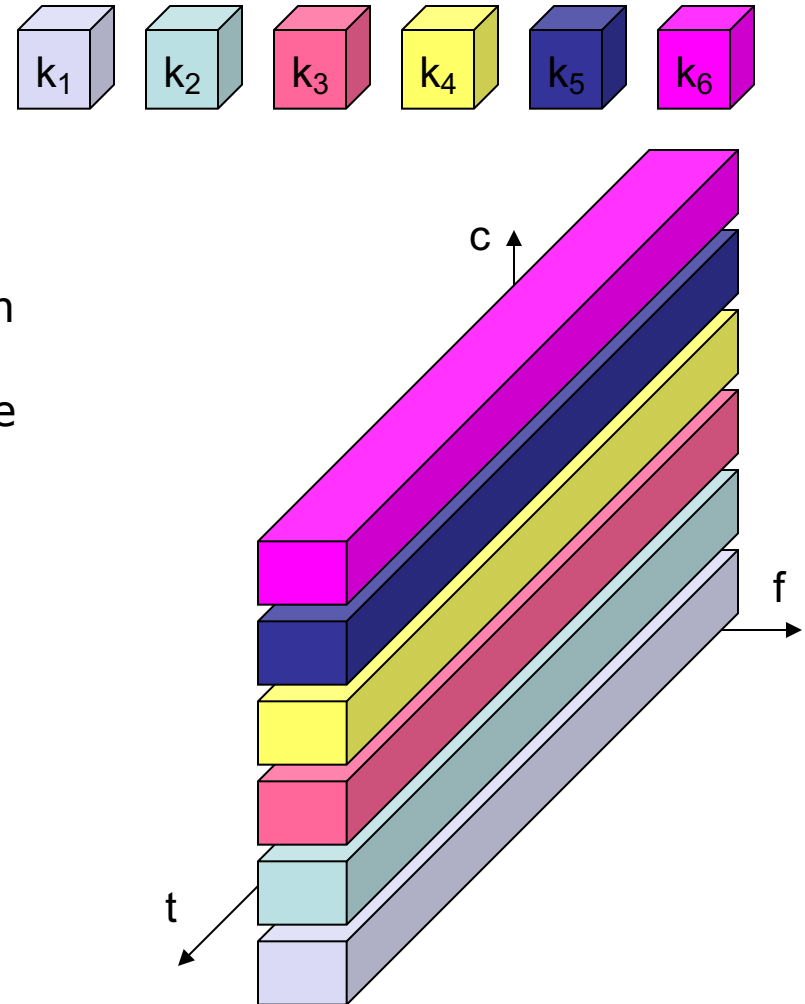
Slotted Aloha

- Slotted Aloha additionally uses time-slots
- All senders have to be synchronized
- Transmission can only start at the beginning of a time slot
- Access is not coordinated
- The introduction of slots raises throughput from 18 to 36%
- Works perfectly under a light load but cannot give any transmission guarantees, such as maximum delay before accessing the medium, or minimum throughput



CDMA

- Each channel has a unique code
- All the channels use the same spectrum at the same time
- Advantages:
 - bandwidth efficient
 - no coordination and synchronization necessary
 - good protection against interference and tapping
- Disadvantages:
 - lower data rates
- Implemented using spread spectrum technology
 - Uses a spreading pseudo-random code
 - The generated code runs at a much higher rate than the data to be transmitted
 - Data for transmission is combined by bitwise **XOR** with the faster code.



CDMA Example

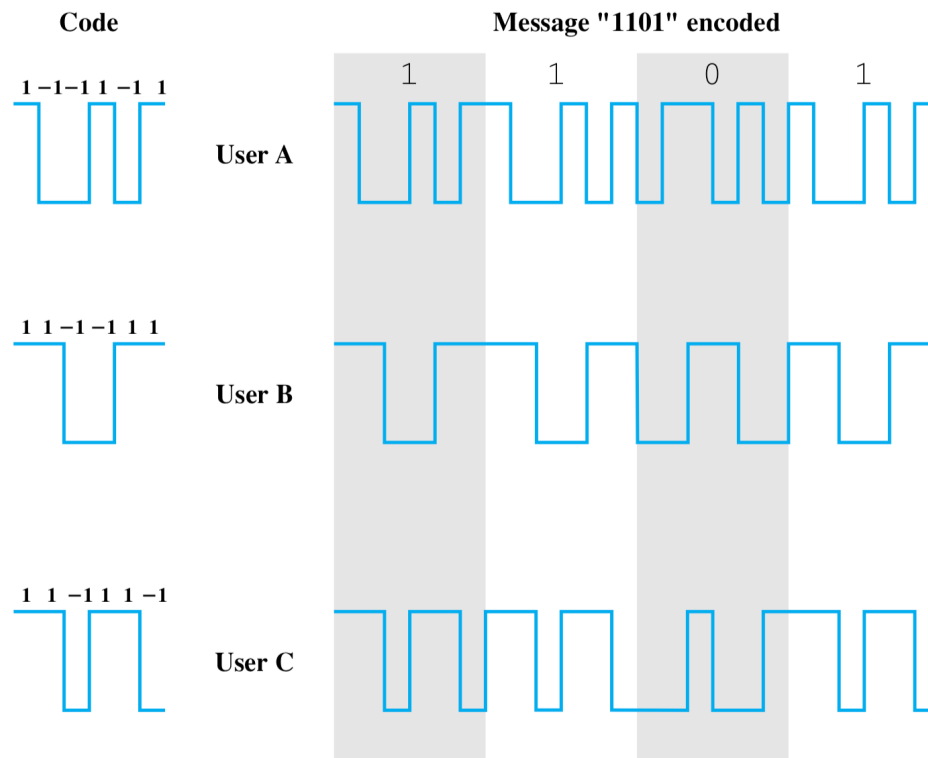
- D = rate of data signal
- Break each bit into k chips
 - Chips are a user-specific fixed pattern
- Chip data rate of new channel = kD
- communication is already synchronized so that the base station knows when to look for codes.
- If $k=6$ and code is a sequence of 1s and -1s
 - For a '1' bit, A sends code as chip pattern
 - $\langle c_1, c_2, c_3, c_4, c_5, c_6 \rangle$
 - For a '0' bit, A sends complement of code
 - $\langle -c_1, -c_2, -c_3, -c_4, -c_5, -c_6 \rangle$
- Receiver knows sender's code and performs electronic decode function

$$S_u(d) = d_1 \times c_1 + d_2 \times c_2 + d_3 \times c_3 + d_4 \times c_4 + d_5 \times c_5 + d_6 \times c_6$$

- $\langle d_1, d_2, d_3, d_4, d_5, d_6 \rangle$ = received chip pattern
- $\langle c_1, c_2, c_3, c_4, c_5, c_6 \rangle$ = sender's code

CDMA Example

- User A code = $\langle 1, -1, -1, 1, -1, 1 \rangle$
 - To send a 1 bit = $\langle 1, -1, -1, 1, -1, 1 \rangle$
 - To send a 0 bit = $\langle -1, 1, 1, -1, 1, -1 \rangle$ (complement)
- User B code = $\langle 1, 1, -1, -1, 1, 1 \rangle$
 - To send a 1 bit = $\langle 1, 1, -1, -1, 1, 1 \rangle$
- User C code = $\langle 1, 1, -1, 1, 1, -1 \rangle$
 - To send a 1 bit = $\langle 1, 1, -1, 1, 1, -1 \rangle$



CDMA Example

- If A sends a 1 bit, then d is $\langle 1, -1, -1, 1, -1, 1 \rangle$ and the preceding computation using $S_u(d)$ with $u=A$ (using A's code) becomes:

$$1 * 1 + (-1) * (-1) + (-1) * (-1) + 1 * 1 + (-1) * (-1) + 1 * 1 = 6$$

- If A sends a 0 bit that corresponds to $d = \langle -1, 1, 1, -1, 1, -1 \rangle$, we get, by always using the A's code:

$$(-1) * 1 + 1 * (-1) + 1 * (-1) + (-1) * 1 + 1 * (-1) + (-1) * 1 = -6$$

- Note that $-6 \leq S_u(d) \leq 6$
- Only d 's resulting in the extreme values of 6 and -6 are A's code and its complement, respectively.
 - if SA produces a +6, we have received a 1 bit from A
 - if SA produces a -6, we have received a 0 bit from A
 - otherwise, someone else is sending information or there is an error.
- If B sends a 1 bit, then $d = \langle 1, 1, -1, -1, 1, 1 \rangle$. Using A's code we obtain:

$$1 * 1 + 1 * (-1) + (-1) * (-1) + (-1) * 1 + 1 * (-1) + 1 * 1 = 0$$

- The unwanted signal (from B) in A does not show up at all.
- if B had sent a 0 bit, the decoder would produce a value of 0 again
- The codes of A and B are **orthogonal**, by having the property:

$$S_A(d_B) = S_B(d_A) = 0$$

CDMA Example

- If codes are not orthogonal, the signal would make a small contribution to the decoded signal instead of 0.
- The CDMA receiver can filter out all the contribution from unwanted users or they appear as low-level noise (additive)
- if there are too many users competing for the channel with the user the receiver is trying to listen to, the system breaks down.
- In (e) we recover the transmission B in the combined signal with a value (8) that adds the noise (2) of the transmission C

(a) User's codes

User A	1	-1	-1	1	-1	1
User B	1	1	-1	-1	1	1
User C	1	1	-1	1	1	-1

(b) Transmission from A

Transmit (data bit = 1)	1	-1	-1	1	-1	1	
Receiver codeword	1	-1	-1	1	-1	1	
Multiplication	1	1	1	1	1	1	= 6

Transmit (data bit = 0)	-1	1	1	-1	1	-1	
Receiver codeword	1	-1	-1	1	-1	1	
Multiplication	-1	-1	-1	-1	-1	-1	= -6

(c) Transmission from B, receiver attempts to recover A's transmission

Transmit (data bit = 1)	1	1	-1	-1	1	1	
Receiver codeword	1	-1	-1	1	-1	1	
Multiplication	1	-1	1	-1	-1	1	= 0

(d) Transmission from C, receiver attempts to recover B's transmission

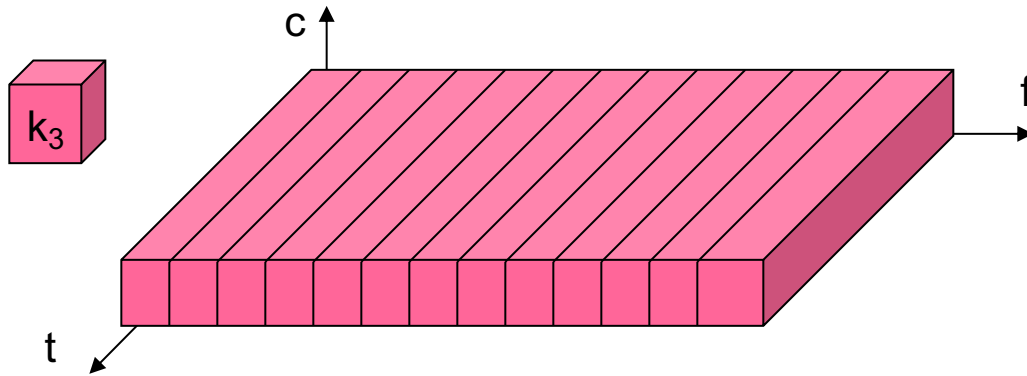
Transmit (data bit = 1)	1	1	-1	1	1	-1	
Receiver codeword	1	1	-1	-1	1	1	
Multiplication	1	1	1	-1	1	-1	= 2

(e) Transmission from B and C, receiver attempts to recover B's transmission

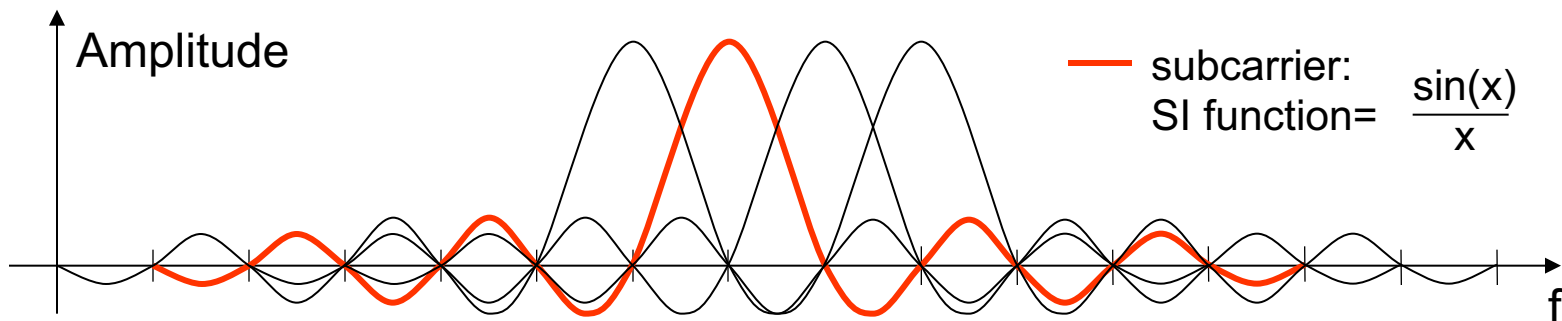
B (data bit = 1)	1	1	-1	-1	1	1	
C (data bit = 1)	1	1	-1	1	1	-1	
Combined signal	2	2	-2	0	2	0	
Receiver codeword	1	1	-1	-1	1	1	
Multiplication	2	2	2	0	2	0	= 8

OFDM (Orthogonal Frequency Division Multiplexing)

- Parallel data transmission on orthogonal subcarriers with lower rate



- Maximum of one subcarrier frequency appears exactly at a frequency where all other subcarriers equal zero
 - superposition of frequencies in the same frequency range

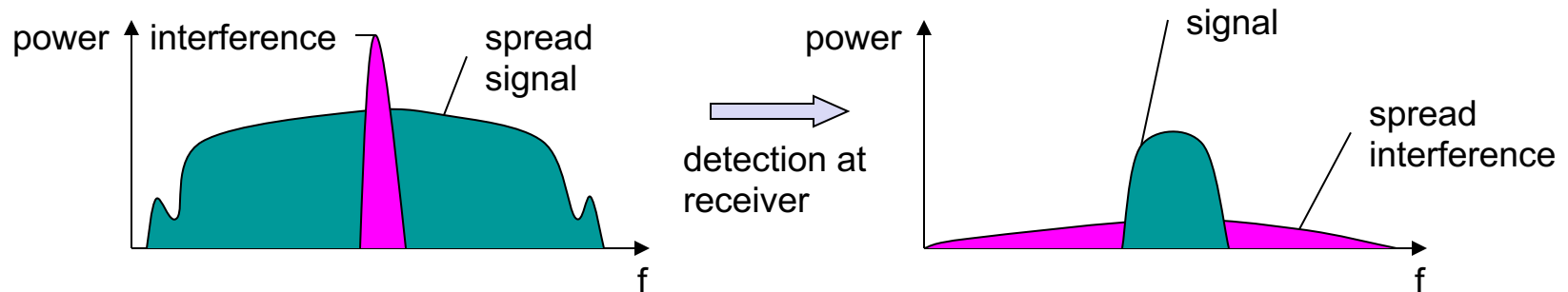


OFDM

- Properties
 - Lower data rate on each subcarrier → less ISI
 - interference on one frequency results in interference of one subcarrier only
 - no guard space necessary
 - orthogonality allows for signal separation via inverse FFT on receiver side
 - precise synchronization necessary (sender/receiver)
- Advantages
 - no equalizer necessary
 - no expensive filters with sharp edges necessary
 - better spectral efficiency (compared to CDM)
- Application
 - 802.11a, HiperLAN2, ADSL

Spread spectrum technology

- Problem of radio transmission: frequency dependent fading can wipe out narrow band signals for duration of the interference
- Solution: spread the narrow band signal into a broad band signal using a special code - protection against narrow band interference

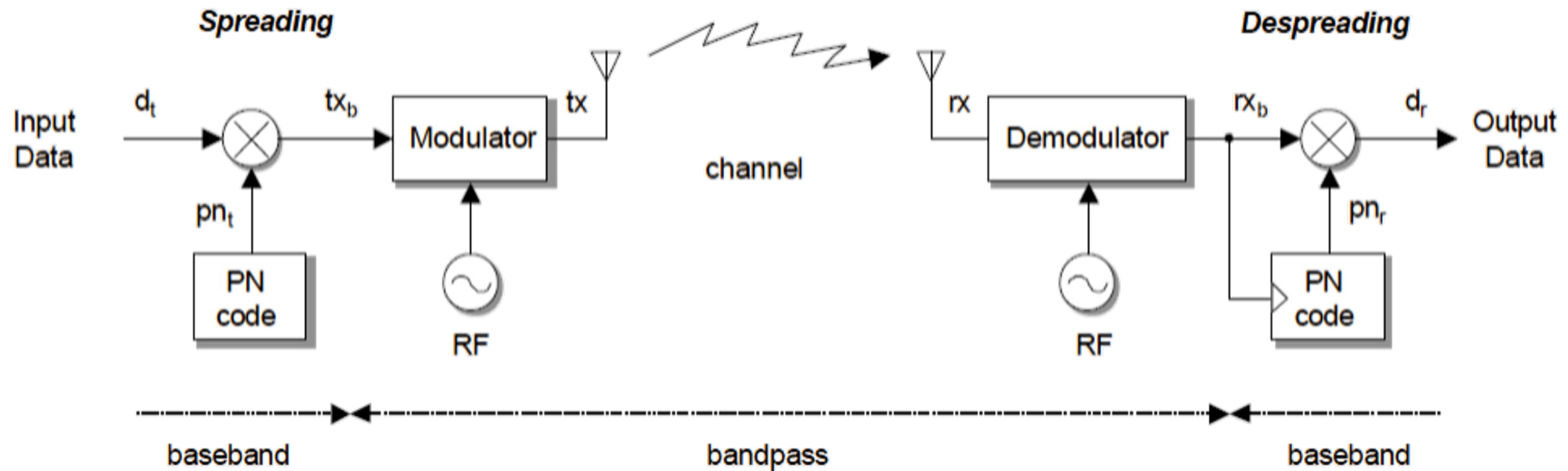


- Side effects:
 - coexistence of several signals without dynamic coordination
 - tap-proof
- Alternatives: Direct Sequence, Frequency Hopping

Spread Spectrum Advantages

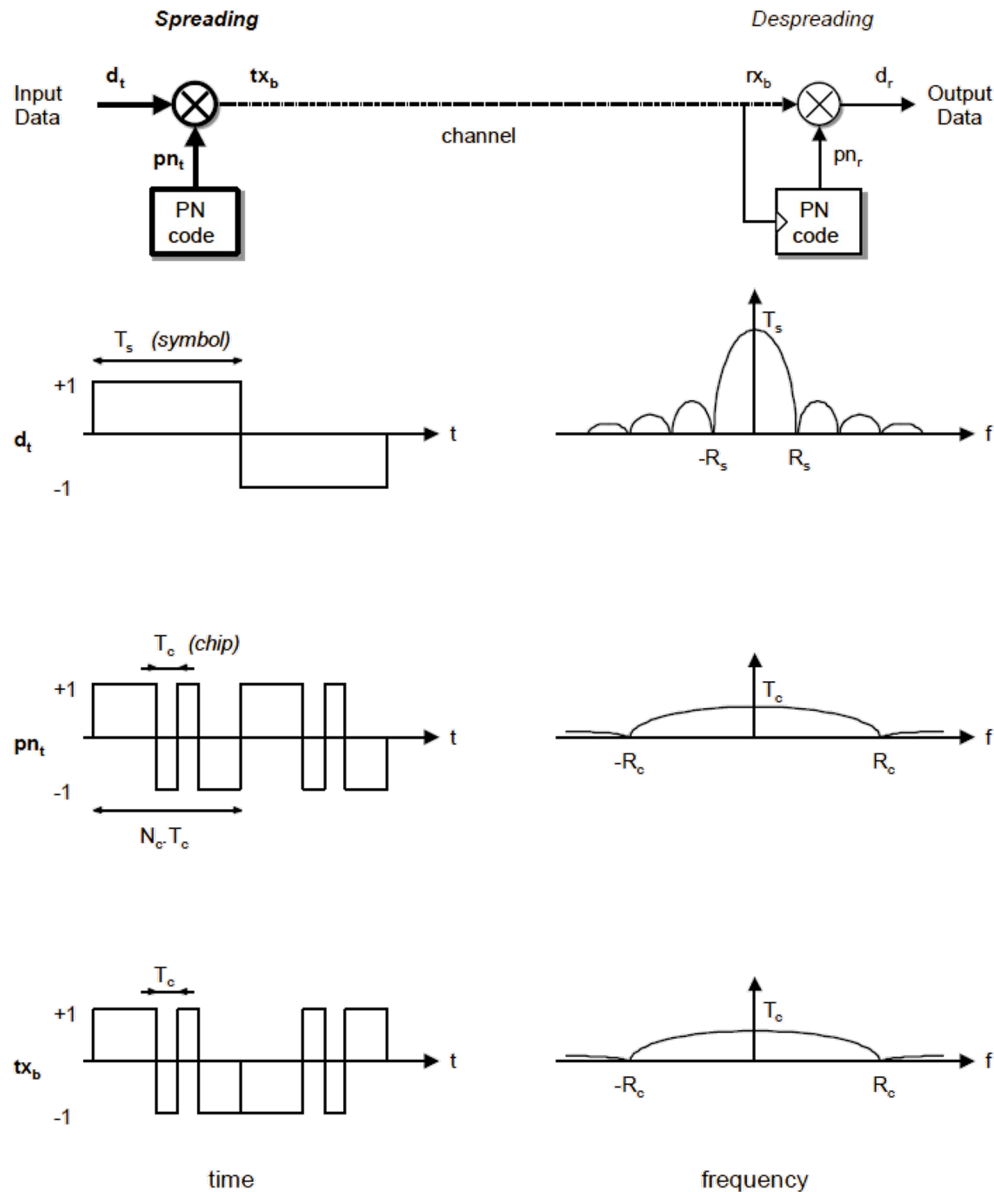
- Hide a signal below the noise floor
- Resistance to narrowband jamming and interference
- Mitigate performance degradation due to intersymbol and narrowband interference
- Spread spectrum modulation is done using a **spreading code** independent of the data in the signal
- Despreading at the receiver is done by correlating the received signal with a synchronized copy of the spreading code

Spread Spectrum block diagram



- Input
 - Binary data d_t with symbol rate $R_s = 1/T_s$
 - Pseudo-noise code: pn_t with chip rate $R_c = 1/T_c$ (an integer of R_s)
- Spreading
 - Transmitted baseband signal: $tx_b = d_t * pn_t$ modulated as tx
- *Despreading*
 - Received modulated signal rx demodulated as rx_b

Spreading

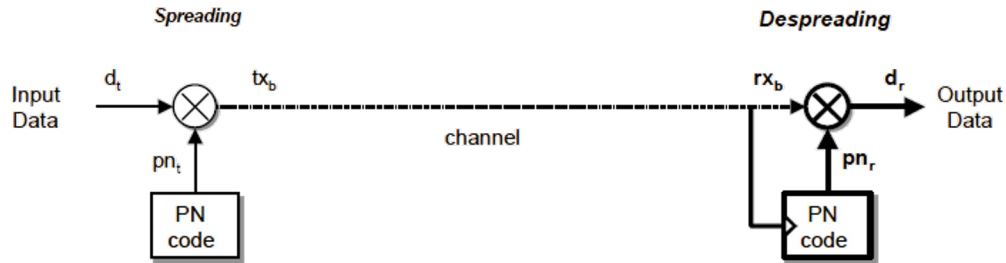


$$BW_{info} \cong R_s \ll BW_{SS} \cong R_c$$

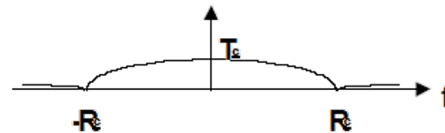
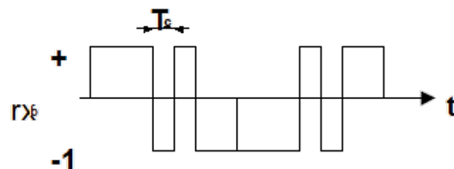
Processing gain

$$SF = G_p = \frac{BW_{SS}}{BW_{info}} = \frac{R_c}{R_s} = \frac{T_b}{T_c} = N_c$$

Despreading



3.2.1 $pn_r = pn_t$



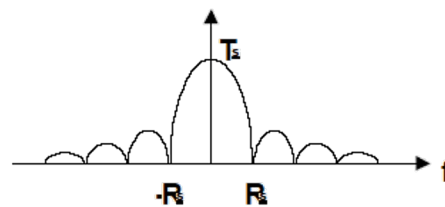
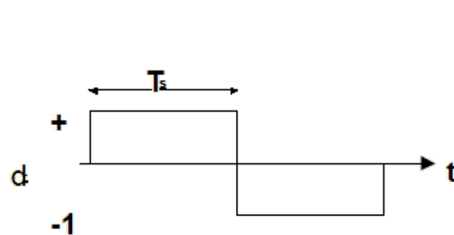
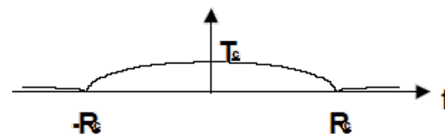
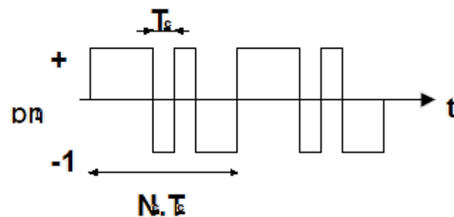
$$d_r = rx_b \cdot pn_r = (d_t \cdot pn_t) \cdot pn_t$$

$$pn_t = +1 +1 +1 -1 +1 -1 -1$$

$$pn_t \cdot pn_t = +1 \quad \text{for all } t$$

$$\text{autocorrelation } Ra(\tau=0) = \text{average}(pn_t \cdot pn_t) = +1$$

$$d_r = d_t$$



$$pn_r \neq pn_t$$

$$d_r = rx_b \cdot pn_r = (d_t \cdot pn_t) \cdot pn_r$$

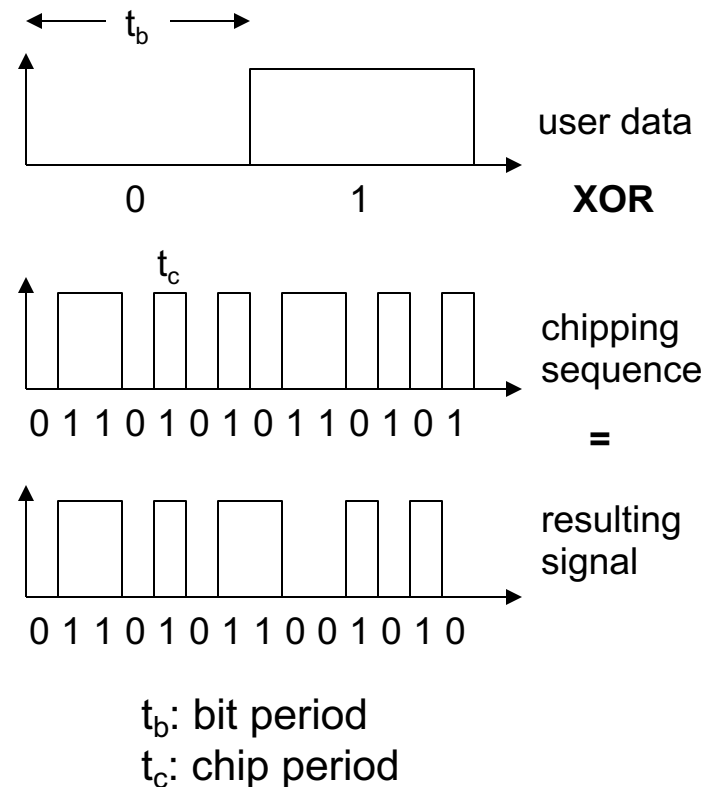
$$\text{crosscorrelation } Rc(\tau) = \text{average}(pn_t \cdot pn_r) \ll 1 \quad \text{for all } \tau$$

time

frequency

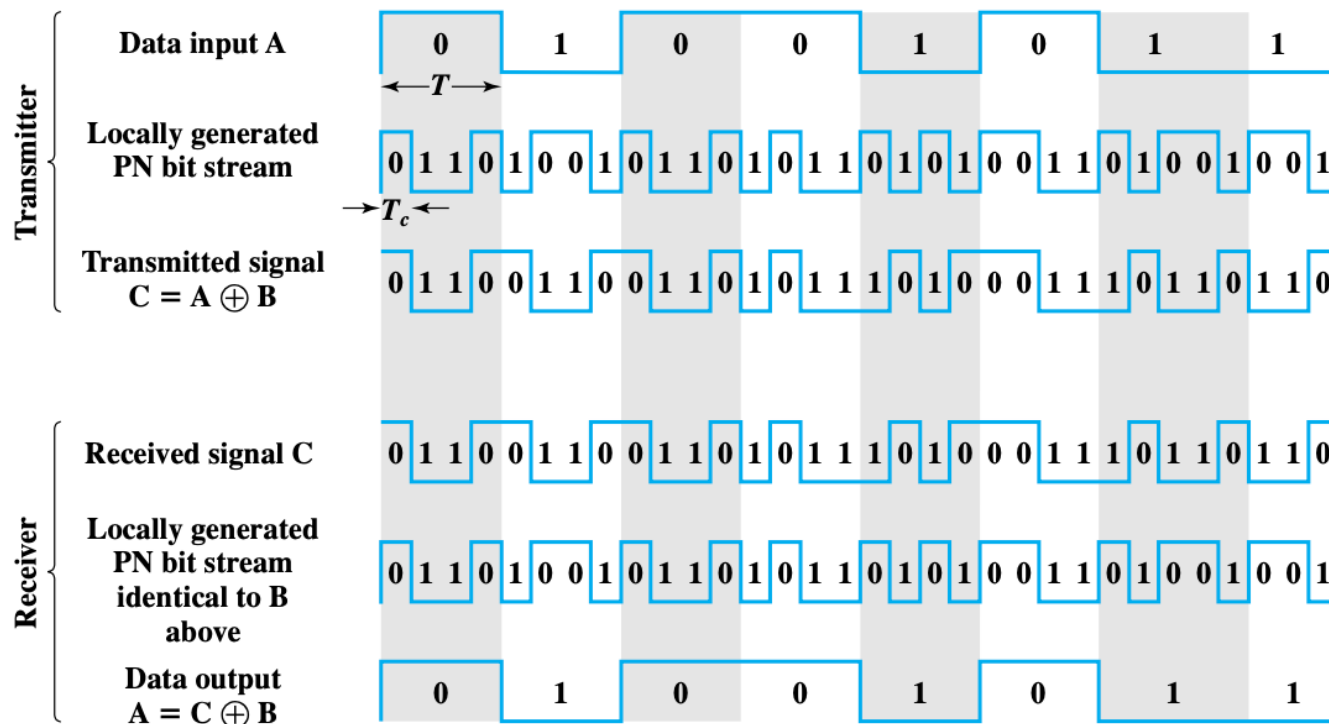
DSSS (Direct Sequence Spread Spectrum)

- XOR of the signal with pseudo-random number pn (chipping sequence)
 - many chips per bit (e.g., 128) result in higher bandwidth of the signal
- Advantages
 - reduces frequency selective fading
 - in cellular networks
 - base stations can use the same frequency range
 - several base stations can detect and recover the signal
 - soft handover
- Disadvantages
 - precise power control necessary



DSSS (Direct Sequence Spread Spectrum)

- Each bit in the original signal is represented by multiple bits in the transmitted signal, using a spreading code.
- The spreading code spreads the signal across a wider frequency band proportional to the number of bits used
- XOR is used to combine the digital information stream with the spreading code bit stream



DSSS (Direct Sequence Spread Spectrum)

- If we use + 1 and - 1 to represent the two binary digits and transmit our signal by using BPSK modulation, it will be:

$$s_d(t) = A d(t) \cos(2\pi f_c t)$$

where:

A = amplitude of signal f_c = carrier frequency

$d(t)$ = discrete function taking for a bit time a value of

- +1 if the corresponding bit in the bit stream is 1 and
- -1 if the corresponding bit in the bit stream is 0

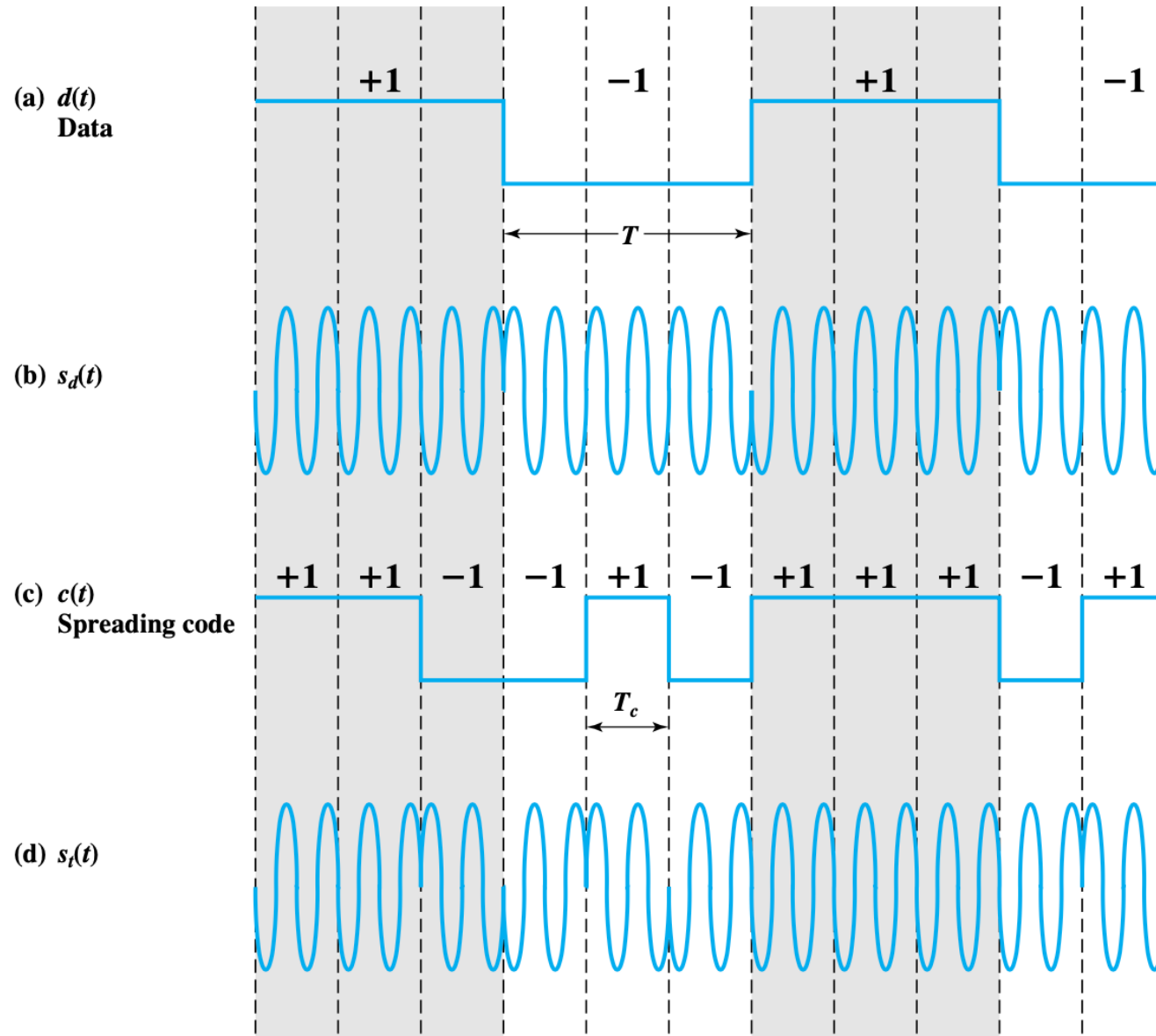
- To produce the DSSS signal, we multiply the $s_d(t)$ by $c(t)$, which is the PN sequence taking on values of +1 and -1:

$$s(t) = A d(t)c(t) \cos(2\pi f_c t)$$

- At the receiver, the incoming signal is multiplied again by $c(t)$. But $c(t)c(t) = 1$ and therefore the original signal is recovered:

$$s(t)c(t) = A d(t)c(t)c(t) \cos(2\pi f_c t) = s_d(t)$$

DSSS (Direct Sequence Spread Spectrum)

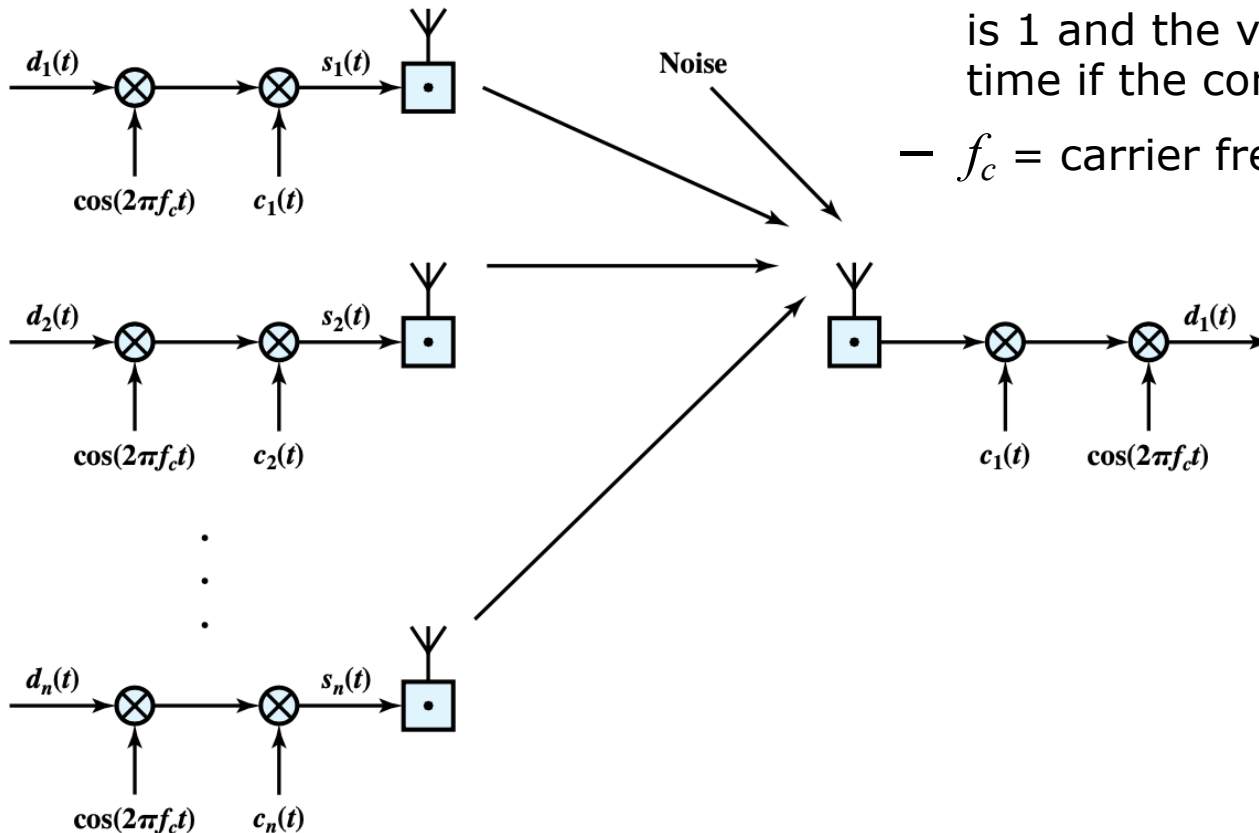


CDMA for DSSS

- n users each using different orthogonal PN sequence
- For each user, the data stream to be transmitted, $d_i(t)$, is BPSK modulated to produce a signal then multiplied by the spreading code for that user, $c_i(t)$
- *The BPSK-modulated signal can be represented as*

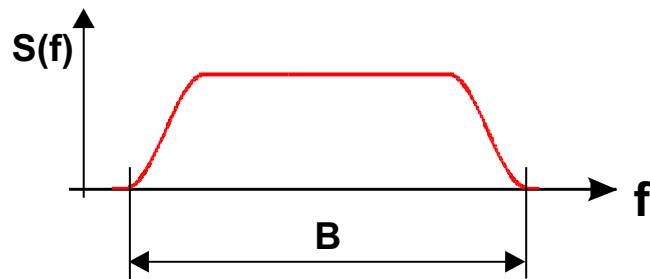
$$s_i(t) = d_i(t) \cos(2\pi f_c t) \quad \text{where:}$$

- $d_i(t)$ = discrete function taking the value of +1 if the corresponding bit is 1 and the value of -1 for one bit time if the corresponding bit is 0
- f_c = carrier frequency

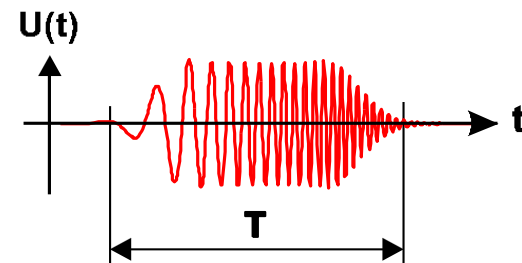


Chirp Spread Spectrum (SS)

- A chirp pulse is a frequency modulated pulse.
 - Its duration is T ; within this time the frequency is changing in a monotonic manner from a lower value to a higher one („Up-Chirp“) or reverse („Down-Chirp“).
 - The difference between these two frequencies is a good approximation for the bandwidth B of the chirp pulse.



Spectrum of the chirp pulse with bandwidth B and a roll-off factor of 0.25



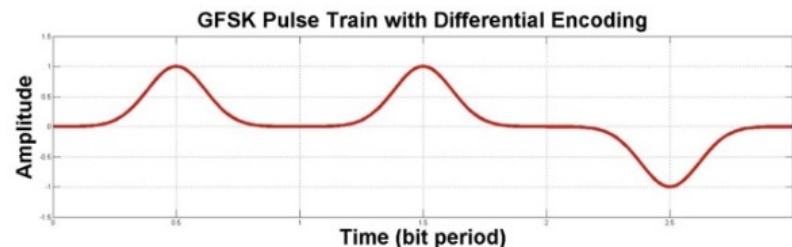
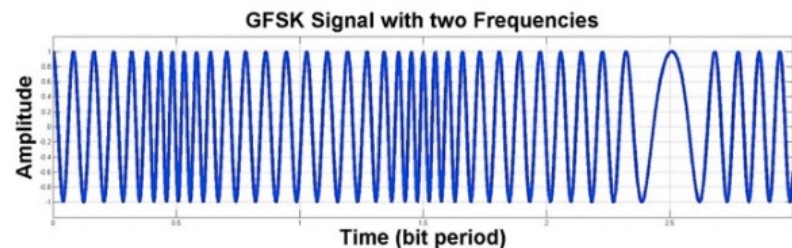
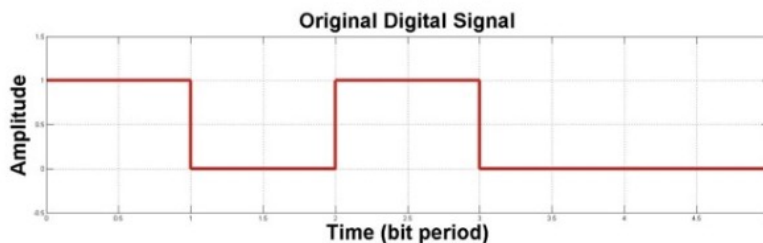
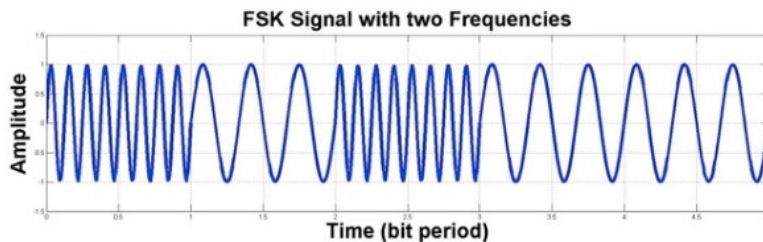
Up-Chirp in the time domain (roll-off factor 0.25)

- Chirp spread spectrum (CSS) is a spread spectrum technique uses wideband linear frequency modulated chirp pulses to encode information
- Because the chirps utilize a broad band of the spectrum, chirp spread spectrum is also resistant to multi-path fading even when operating at very low power.

Gaussian FSK modulation

- is an extension of the FSK modulation scheme where the frequency of the modulated signal will not instantaneously change at the beginning of each symbol period of the binary data.
- a Gaussian filter is included before the baseband waveform signal, as a result, the transition from bit 0 to bit 1 becomes smoother.
- the amplitude and phase variations of the modulated signal will be relatively less compared to traditional FSK

Gaussian Filter



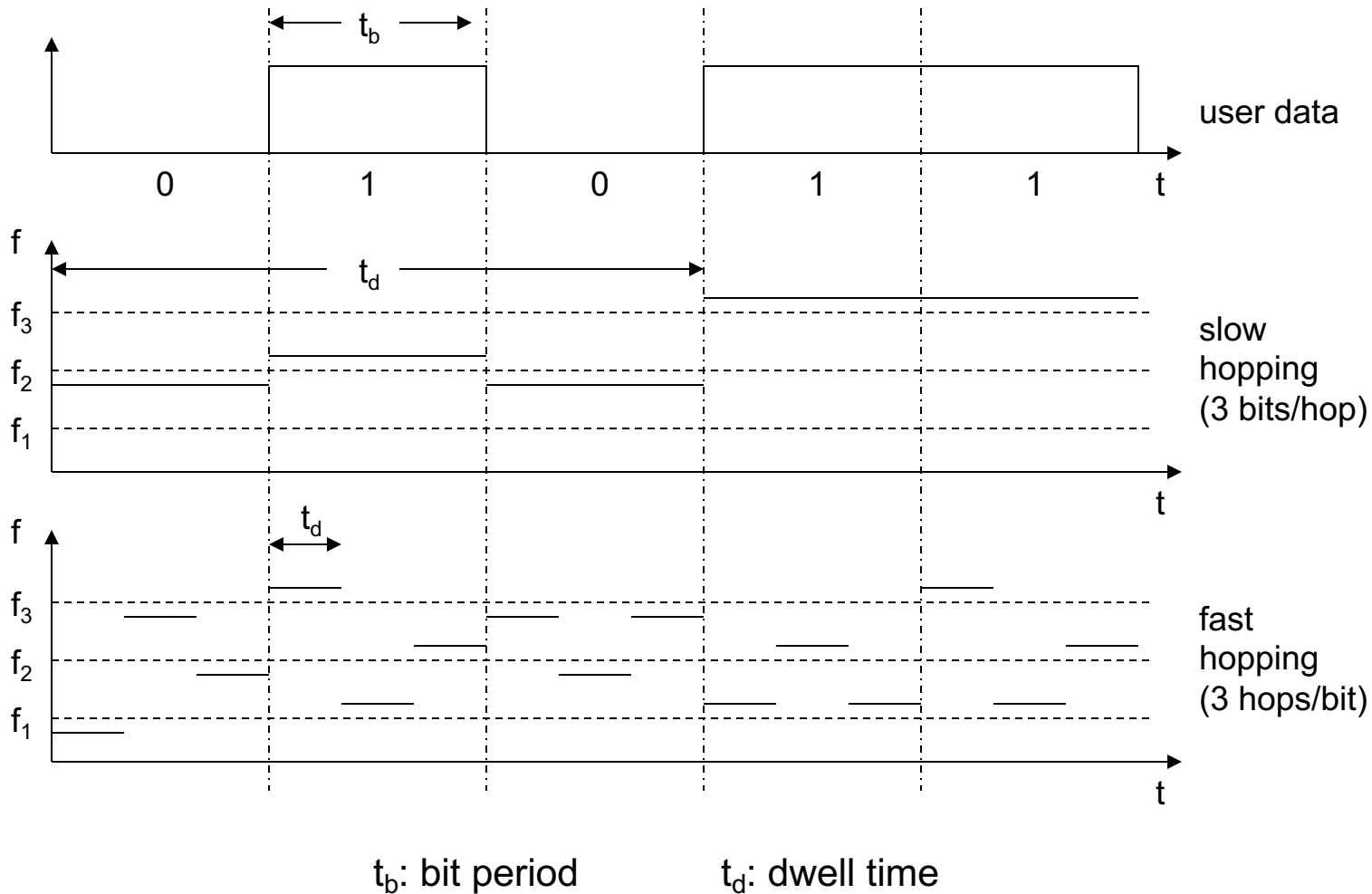
Frequency Hopping Spread Spectrum (FHSS)

- Signal is broadcast over seemingly random series of radio frequencies
- Signal hops from frequency to frequency at fixed intervals
- Channel sequence dictated by spreading code
- Receiver, hopping between frequencies in synchronization with transmitter, picks up message
- Advantages
 - Eavesdroppers hear only unintelligible blips
 - Attempts to jam signal on one frequency succeed only at knocking out a few bits

FHSS (Frequency Hopping)

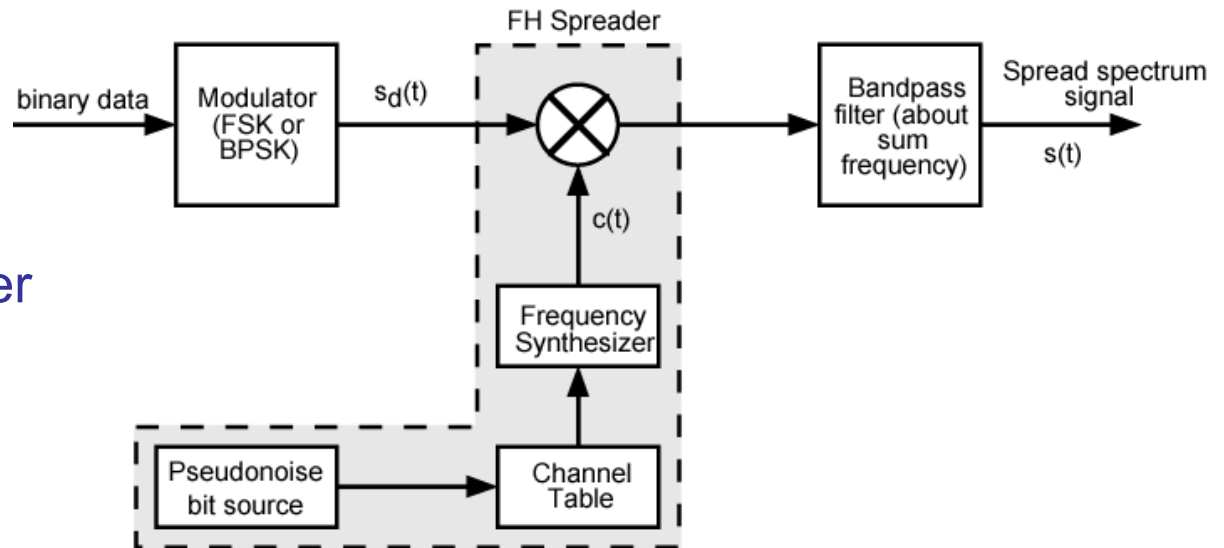
- Discrete changes of carrier frequency
 - sequence of frequency changes determined via pseudo random number sequence
- Two versions
 - Fast Hopping: several frequencies per user bit
 - Slow Hopping: several user bits per frequency
- Advantages
 - frequency selective fading and interference limited to short period
 - simple implementation
 - uses only small portion of spectrum at any time
- Disadvantages
 - not as robust as DSSS
 - simpler to detect

Slow and Fast FHSS



FHSS Block Diagram

Transmitter



Receiver

